



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 7 • STANDARD 6.AT.8

Write Equations

Flagship

Lesson 7-1-flagship • Teacher Copy

Name: _____

Date: _____

Class: _____

COLD CASE MISSION

| The Equation Vault

You are Detective Ruiz's new code analyst. A tip just came in: 'A number of stolen gems plus 8 more equals 20 total.' The evidence vault only opens when the clue is translated into a correct equation. Master writing equations and the case cracks wide open.



TODAY'S OBJECTIVES

Content Objective: I can write an equation to represent a real-world situation.

Language Objective: I can explain my equation using the words equation, variable, equal sign, and expression.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Equation

Variable

Equal sign

Expression



Tap any word to see what it means and a picture.

1

A math sentence that says two amounts are equal, with an = sign, is an

2

A letter that stands for an unknown number is a .

3

The symbol = that shows two amounts are the same is the

4

A group of numbers and variables with no equal sign is an

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

On the balance scale you put $n + 8$ on one side and 20 on the other. How did you decide what belongs on each side of the equal sign?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Equation** — A math sentence with an equal sign showing both sides are the same.
Ecuación — Una oración matemática con un signo igual que muestra que ambos lados son iguales.

☐ **Variable** — A letter that stands for an unknown number.
Variable — Una letra que representa un número desconocido.

☐ **Equal sign** — The = sign, showing both sides are the same.
Signo igual — El signo =, que muestra que ambos lados son iguales.

☐ **Expression** — A math phrase with numbers and letters, but no equal sign.
Expresión — Una frase matemática con números y letras, pero sin signo igual.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

I put _____ on the left side because _____.

Puse _____ en el lado izquierdo porque _____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: I let a variable stand for the unknown number of gems.

Evidence: Students choose a variable (such as n) for the unknown gems, build $n + 8 = 20$, and connect 'plus 8 more' to $+ 8$ and 'equals 20 total' to $= 20$.

Reasoning: This evidence connects the definition of equation to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **I put ____ on the left side because ____.**
Puse ____ en el lado izquierdo porque ____.
 - **The equal sign means both sides are ____.**
El signo igual significa que ambos lados son ____.
-
-
-
-

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
 - **I know because ____.**
Lo sé porque ____.
 - **So ____.**
Entonces ____.
-
-
-
-

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Equation
2. **Notes 2:** Variable
3. **Notes 3:** Equal sign
4. **Notes 4:** Expression
5. **Watch for this mistake:** *A common mistake in Write Equations is reversing the order in 'less than' and 'fewer than' clues — writing $7 - n = 18$ instead of $n - 7 = 18$ for 'seven less than a number is 18.' These phrases flip the words, but you still start with the unknown number and subtract from it. Before you submit, plug in your answer to check: for $n - 7 = 18$, $n = 25$ works because $25 - 7 = 18$ ✓, but $7 - 25 = -18$ ✗ shows the reversed equation is wrong.*
6. **Try It 1:** A. $6c = 54$ — *Let c be the number of stolen coins. 'Six times' means multiply by 6, so the equation is $6c = 54$.*
7. **Try It 2:** A. $b / 5 = 4$ — *Let b be the total gold bars. Sharing equally into 5 cases means $b / 5$, and each case has 4, so $b / 5 = 4$.*